Equality in the Demailly–Pham Inequality: An Initial Proof Draft

Conjecture 6.2 and the divisorial case of Conjecture 6.3.

By GPT-6 Astra, September 8, 2026

I believe the following proves Conjecture 6.2 in full. It proves the divisorial case of the corrected Conjecture 6.3, which is sufficient for Conjecture 6.2. I do **not** claim here to prove the stronger assertion for every real valuation of finite log discrepancy.

The key point is to show that, in the equality case, restriction of the valuation filtration to a general hyperplane is again a valuation filtration. A result of Zhiyuan Chen supplies the geometric rigidity needed for that step.

1. The statement proved

Let

$$R = \mathbb{C}[\![x_1, \dots, x_n]\!], \qquad E_l(I) = \sum_{j=1}^{l} \frac{\sigma_{j-1}(I)}{\sigma_j(I)}, \qquad \sigma_0(I) = 1.$$

We prove simultaneously, by induction on n, the following two assertions.

Ideal assertion P_n. If $\mathrm{ht}(I) \geq l$ and

$$\mathrm{lct}(I) = E_l(I),$$

then there are regular parameters $z_1, \dots, z_n$ and positive integers $d_1, \dots, d_l$ such that

$$\boxed{\overline{I} = \overline{(z_1^{d_1}, \dots, z_l^{d_l})}.}$$

Together with the known converse, this is Conjecture 6.2. arXiv

Divisorial assertion D_n. If v is a positive real multiple of a divisorial valuation, its center has height at least l, and

$$E_l(\mathfrak{a}_\bullet(v)) = A_R(v),$$

then v is monomial in regular parameters, with exactly l positive weights.

The induction will have the form

$$P_{n-1} \Longrightarrow D_n \Longrightarrow P_n.$$

We use the paper's inequality

$$E_l(\mathfrak{a}_\bullet(v)) \leq \mathrm{lct}(\mathfrak{a}_\bullet(v)) \leq A_R(v). \tag{1}$$

We also use the following numerical consequence of Bivià-Ausina's Proposition 10: if $I \subseteq J$ have height at least l, then $E_l(I) \leq E_l(J)$, and equality forces

$$\sigma_j(I) = \sigma_j(J) \qquad (1 \leq j \leq l). \tag{2}$$

For non-primary ideals, this follows by taking a common general l-dimensional section. arXiv

All birational constructions below are over $\operatorname{Spec} R$. The relative MMP and vanishing inputs are available over excellent characteristic-zero local bases; Chen explicitly includes this setting in his Remark 1.5. Thus the argument does not require algebraizing the formal parameters. arXiv

2. A hyperplane identity for finitely generated valuation filtrations

Here is the numerical ingredient.

Lemma. Suppose $v = \operatorname{ord}_E$ is centered at the maximal ideal, and its valuation algebra is finitely generated. Then

$$\boxed{\frac{e_{n-1}(\mathfrak{a}_\bullet(v))}{e_n(\mathfrak{a}_\bullet(v))} = v(\mathfrak{m}).} \tag{3}$$

Proof. Write $\mathfrak{a}_m = \mathfrak{a}_m(v)$. Choose $q > 0$ sufficiently divisible that, with $I = \mathfrak{a}_q$,

$$\mathfrak{a}_{qk} = I^k \qquad (k \geq 0).$$

Consequently,

$$e_j(\mathfrak{a}_\bullet) = \frac{e_j(I)}{q^j}.$$

Choose a general linear form h, put

$$c = v(h) = v(\mathfrak{m}), \qquad S = R/(h), \qquad \mathfrak{b}_m = \mathfrak{a}_m S.$$

Generic-section multiplicities give

$$e(IS; S) = e_{n-1}(I; R).$$

This is the generic-section formula recorded in the paper's Propositions 2.35 and 2.37. arXiv

Finite generation and the Hilbert–Samuel polynomial now give

$$\ell_R(R/\mathfrak{a}_m) = \frac{e_n(\mathfrak{a}_\bullet)}{n!} m^n + o(m^n),$$

and

$$\ell_S(S/\mathfrak{b}_m) = \frac{e_{n-1}(\mathfrak{a}_\bullet)}{(n-1)!} m^{n-1} + o(m^{n-1}).$$

Indeed, these statements first hold along $m = qk$, where the ideals are powers of fixed ideals; monotonicity gives them for all m.

The valuation property gives the exact colon identity

$$(\mathfrak{a}_m : h) = \mathfrak{a}_{m-c}.$$

Hence

$$0 \longrightarrow R/\mathfrak{a}_{m-c} \xrightarrow{\cdot h} R/\mathfrak{a}_m \longrightarrow S/\mathfrak{b}_m \longrightarrow 0.$$

Summing along multiples of c, rather than differentiating an asymptotic formula, gives

$$\ell_R(R/\mathfrak{a}_{cM}) = \sum_{k=1}^{M} \ell_S(S/\mathfrak{b}_{ck}).$$

Comparing leading coefficients yields

$$c^n e_n(\mathfrak{a}_\bullet) = c^{n-1} e_{n-1}(\mathfrak{a}_\bullet),$$

which proves (3). □

In particular, after a general hyperplane restriction,

$$E_{n-1}(\mathfrak{b}_\bullet) = E_n(\mathfrak{a}_\bullet) - v(\mathfrak{m}). \tag{4}$$

The remaining problem is that $\mathfrak{b}_\bullet$ need not initially be a valuation filtration.

3. The geometric input that repairs restriction

We need the following specialization of Chen's Theorem 3.5.

LC-intersection lemma. Let $\mu : Y \to X$ be proper, with X affine. Suppose E, T are distinct reduced $\mathbb{Q}$-Cartier divisors such that

$$(Y, E+T) \text{ is lc}, \qquad -(K_Y + E + T) \text{ is } \mu\text{-ample},$$

every lc center is contained in $E \cup T$, and $E \cap T$ is irreducible. Then

$$\bigoplus_{m \geq 0} \frac{H^0(Y, \mathcal{O}_Y(-mE))}{H^0(Y, \mathcal{O}_Y(-(m+1)E)) + H^0(Y, \mathcal{O}_Y(-mE - T))} \tag{5}$$

is an integral domain.

To obtain this from Chen, take the section-ring divisor $L_1 = -E$ and the two boundary components E, T. The algebra (5) is the subalgebra of his central extended-Rees quotient with boundary multidegree $(0, 0)$. His Theorem 3.5(3) makes that quotient a domain when the intersection is irreducible. arXiv

The importance of the lemma is that it controls **all graded pieces**, not merely a sufficiently divisible Veronese.

4. Proof of the divisorial assertion D_n

Assume P_{n-1}. Rescaling v, we may take

$$v = \mathrm{ord}_E\,.$$

Write

$$A = A_R(v), \qquad \mathfrak{a}_m = \mathfrak{a}_m(v).$$

4.1. Equality gives a finitely generated extraction

By (1),

$$\mathrm{lct}(\mathfrak{a}_\bullet) = A.$$

Blum's Theorem 1.3 and Proposition 1.5 give finite generation. If the center has codimension at least two, the resulting model

$$\mu : Y = \mathrm{Proj}_X \bigoplus_{m\geq 0} \mathfrak{a}_m \longrightarrow X$$

has a unique exceptional prime E, with $-E$ relatively ample, Y klt, and (Y, E) lc. Moreover,

$$K_Y = \mu^* K_X + (A-1)E. \tag{6}$$

The extraction argument applies over our excellent local base using the relative MMP mentioned above. arXiv

Choose q sufficiently divisible that

$$I := \mathfrak{a}_q, \qquad \mathfrak{a}_{qk} = I^k, \qquad I\mathcal{O}_Y = \mathcal{O}_Y(-qE). \tag{7}$$

In particular,

$$v(I) = q, \qquad qE_l(I) = q\,\mathrm{lct}(I) = A. \tag{8}$$

The center has height **exactly** l. Indeed, if its height were at least $l+1$, finite generation would give positive finite asymptotic multiplicities through index $l+1$, and therefore

$$E_{l+1}(\mathfrak{a}_\bullet) > E_l(\mathfrak{a}_\bullet) = A,$$

contradicting (1).

If $l = 1$, the center is a principal prime (f), and the primitive valuation is ord_f. Then

$$A_R(v) = 1, \qquad E_1(\mathfrak{a}_\bullet(v)) = \frac{1}{\mathrm{ord}_{\mathfrak{m}}(f)}.$$

Equality forces f to be a parameter. Thus we may assume $l \geq 2$, so the extraction above applies.

4.2. Restriction attains equality in dimension $n-1$

Choose a general linear form h, and set

$$S = R/(h), \qquad H = \operatorname{Spec} S, \qquad J = IS, \qquad \mathfrak{b}_m = \mathfrak{a}_m S.$$

Put

$$c = v(h), \qquad r = \min(l, n-1).$$

There are two cases.

When $l = n$, the center is the closed point, $c = v(\mathfrak{m}) > 0$, and the hyperplane identity gives

$$E_r(\mathfrak{b}_\bullet) = A - c.$$

When $l < n$, choose h outside the center prime. Then $c = 0$, and generic restriction preserves $\sigma_1, \dots, \sigma_l$, giving the same formula.

Since $\mathfrak{b}_{qk} = J^k$, both cases yield

$$\boxed{E_r(J) = \frac{A-c}{q}.} \tag{9}$$

Let $t = \operatorname{lct}_S(J)$. We claim that

$$t \le \frac{A-c}{q}.$$

For $s < t$, the pair (H, J^s) is klt. Klt inversion of adjunction gives that $(X, H+I^s)$ is plt along H. The excellent-local version is available in Sato–Takagi, Theorem 3.2. Letting $s \uparrow t$ gives

$$(X, H + I^t) \text{ is lc.}$$

Here a neighborhood of H is all of X, because X is local and H contains its closed point. arXiv

Evaluating the lc inequality at E gives

$$A - c - tq \ge 0.$$

On the other hand, the Demailly–Pham inequality and (9) give

$$t \ge E_r(J) = \frac{A-c}{q}.$$

Thus

$$\boxed{\operatorname{lct}_S(J) = E_r(J) = \frac{A-c}{q}.} \tag{10}$$

By P_{n-1}, there are regular parameters $y_1, \dots, y_{n-1}$ of S and positive integers $d_1, \dots, d_r$ such that

$$\overline{J} = \overline{D}, \qquad D = (y_1^{d_1}, \dots, y_r^{d_r}). \tag{11}$$

4.3. The exceptional divisor meets the strict transform irreducibly

Let $T \subset Y$ be the strict transform of H. Then

$$\mathrm{div}_Y(h) = T + cE.$$

Using (6), (7), and $tq = A - c$, the lc statement above becomes

$$(Y, E + T) \text{ is lc.} \tag{12}$$

Furthermore,

$$-(K_Y + E + T) \sim_{\mu,\mathbb{Q}} -(A - c)E,$$

which is relatively ample because $A - c = qE_r(J) > 0$.

Both E and T are $\mathbb{Q}$-Cartier, and every lc center of this pair lies in $E \cup T$, since Y is klt.

It remains to check that

$$F := E \cap T$$

is irreducible.

By (7), $Y = \mathrm{Bl}_I X$. The quotient of Rees algebras

$$\bigoplus_{k\geq 0} I^k \twoheadrightarrow \bigoplus_{k\geq 0} J^k$$

identifies

$$T = \mathrm{Bl}_J H.$$

Indeed, the latter is integral and agrees with H away from the center, so its image is exactly the strict transform.

By (11), the normalized blowup of J is the normalized blowup of the pure-power ideal D. That normalized blowup has exactly one prime divisor in the support of the pulled-back ideal: its Newton polyhedron has the single noncoordinate facet

$$\frac{\alpha_1}{d_1} + \cdots + \frac{\alpha_r}{d_r} = 1.$$

This also covers $r = 1$, when the support is simply the corresponding coordinate divisor.

Normalization is finite and surjective. Therefore the support of $J\mathcal{O}_T$ is irreducible. But

$$J\mathcal{O}_T = (I\mathcal{O}_Y)|_T = \mathcal{O}_Y(-qE)|_T,$$

whose vanishing support is $E \cap T$. Hence F is irreducible.

4.4. The restricted associated graded ring is a domain

We can now apply the LC-intersection lemma.

Since E is exceptional over a normal base,

$$H^0(Y, \mathcal{O}_Y(-mE)) = \mathfrak{a}_m$$

for $m \geq 0$; extend the notation by $\mathfrak{a}_j = R$ for $j \leq 0$. The identity

$$T = \mathrm{div}_Y(h) - cE$$

gives

$$H^0(Y, \mathcal{O}_Y(-mE - T)) = h\mathfrak{a}_{m-c}.$$

Thus (5) becomes

$$\bigoplus_{m \geq 0} \frac{\mathfrak{a}_m}{\mathfrak{a}_{m+1} + h\mathfrak{a}_{m-c}}.$$

But

$$\mathfrak{a}_m \cap hR = h\mathfrak{a}_{m-c},$$

so this is precisely

$$\boxed{\mathrm{gr}_{\mathfrak{b}_\bullet}(S) = \bigoplus_{m \geq 0} \mathfrak{b}_m/\mathfrak{b}_{m+1}.}$$

Consequently,

$$\mathrm{gr}_{\mathfrak{b}_\bullet}(S) \quad \text{is an integral domain.} \tag{13}$$

This is the step that turns the restricted filtration into an actual valuation filtration.

4.5. The restricted valuation is monomial

The filtration $\mathfrak{b}_\bullet$ is separated: its q-th Veronese consists of powers of the proper ideal J. Hence

$$w(f) = \max\{m : f \in \mathfrak{b}_m\}$$

is finite for every nonzero $f \in S$.

Equation (13) implies

$$w(fg) = w(f) + w(g).$$

Thus w is a valuation and

$$\mathfrak{b}_m = \mathfrak{a}_m(w).$$

In particular, every

$$J^k = \mathfrak{b}_{qk}$$

is integrally closed. Combining this with (11),

$$J^k = \overline{D^k}. \tag{14}$$

Let ν be the monomial valuation on S with weights

$$\nu(y_i) = \frac{1}{d_i} \quad (1 \le i \le r), \qquad \nu(y_i) = 0 \quad (i > r).$$

The Newton description of $\overline{D^k}$ says that

$$\overline{D^k} = \{f : \nu(f) \ge k\}.$$

Therefore (14) gives, for all positive integers k,

$$w(f) \ge qk \quad \Longleftrightarrow \quad \nu(f) \ge k.$$

Apply this equivalence to f^N, and let $N \to \infty$. It follows that

$$\boxed{w = q\nu.} \tag{15}$$

Thus the restricted valuation is monomial, not merely asymptotically equivalent to a monomial filtration.

4.6. Lift the parameters and the whole valuation filtration

For each y_i, choose a lift $z_i \in R$ with

$$v(z_i) = w(y_i).$$

Such a lift exists by the definition of $\mathfrak{b}_m$; a lift of larger value would contradict the definition of $w(y_i)$.

Then

$$h, z_1, \ldots, z_{n-1}$$

is a regular system of parameters for R.

Let M_m be the monomial ideal in these parameters defined by the weights

$$c, w(y_1), \ldots, w(y_{n-1}).$$

Certainly

$$M_m \subseteq \mathfrak{a}_m.$$

Their images modulo h agree, by (15). Using the colon identity once more gives

$$\mathfrak{a}_m = M_m + h\mathfrak{a}_{m-c}. \tag{16}$$

If $c > 0$, induction on m in (16), starting with $\mathfrak{a}_j = M_j = R$ for $j \le 0$, proves

$$\mathfrak{a}_m = M_m \qquad \text{for all } m.$$

If $c = 0$, equation (16) says that the finitely generated module

$$\mathfrak{a}_m/M_m$$

equals its product with h. Nakayama's lemma again gives

$$\mathfrak{a}_m = M_m.$$

Therefore v is monomial in the lifted regular parameters. There are exactly l positive weights: n when $l = n$, and the l positive weights of w when $l < n$.

This proves D_n.

5. From D_n to Conjecture 6.2

Now let $K \subset R$ have height at least l, with

$$\operatorname{lct}(K) = E_l(K).$$

Choose a divisorial valuation computing its log canonical threshold, and rescale it to a valuation u satisfying

$$u(K) = 1, \qquad A_R(u) = \operatorname{lct}(K).$$

Since

$$K^m \subseteq \mathfrak{a}_m(u),$$

we obtain

$$E_l(K) \leq E_l(\mathfrak{a}_\bullet(u)) \leq \operatorname{lct}(\mathfrak{a}_\bullet(u)) \leq A_R(u) = E_l(K).$$

Thus equality holds throughout.

By D_n, u is monomial in regular parameters, with exactly l positive weights. Set

$$M = \mathfrak{a}_1(u).$$

Then M is an integrally closed monomial ideal, involving only those l positive-weight parameters, and

$$K \subseteq M.$$

Furthermore,

$$E_l(K) \leq E_l(M) \leq \operatorname{lct}(M) \leq \frac{A_R(u)}{u(M)} \leq A_R(u) = E_l(K).$$

Consequently,

$$E_l(M) = \operatorname{lct}(M) = E_l(K). \tag{17}$$

We now use only the **already established monomial case**, not the induction assertion in dimension n. Bivià-Ausina's Theorem 13 implies that a monomial ideal attaining the Demailly–Pham bound is diagonal. Applied in the l positive-weight variables, it gives

$$M = \overline{(z_1^{d_1}, \ldots, z_l^{d_l})}$$

for positive integers d_i. arXiv

By (2) and (17),

$$\sigma_l(K) = \sigma_l(M).$$

Also, M has analytic spread l, because it is the integral closure of an ideal generated by l elements. Both ideals have height l: the center of u has height l and contains K.

The paper's Lemma 6.6 therefore applies to $K \subseteq M$, giving

$$\boxed{\overline{K} = M = \overline{(z_1^{d_1}, \ldots, z_l^{d_l})}.}$$

This proves P_n. arXiv

The base case $n = 1$ is immediate because every nonzero proper ideal in $\mathbb{C}[[x]]$ is a power of (x). The simultaneous induction is complete.

6. Exact scope of the conclusion

The argument gives

$$\boxed{\text{Conjecture 6.2 in every dimension and codimension,}}$$

and

$$\boxed{\text{Conjecture 6.3 for positive multiples of divisorial valuations.}}$$

The essential sequence is

$$\begin{aligned}\text{equality} &\Longrightarrow \text{finitely generated extraction} \Longrightarrow \text{equality on a hyperplane} \\ &\Longrightarrow \text{irreducible lc intersection} \Longrightarrow \text{restricted graded ring is a domain} \Longrightarrow \text{monomiality.}\end{aligned}$$

For an arbitrary finite-discrepancy real valuation, the finite-generation step is not supplied by this argument. Thus the full corrected Conjecture 6.3 remains outside what I have proved here, but **the ideal-classification conjecture does not require that additional generality**.